

UNIT-2: Magnetostatic Fields

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Chapter 1

Magnetostatic Fields

We now focus our attention on static magnetic fields, which are characterized by $\mathbf{H}$ or $\mathbf{B}$. There are similarities and dissimilarities between electric and magnetic fields. As $\mathbf{E}$ and $\mathbf{D}$ are related according to $\mathbf{D} = \epsilon\mathbf{E}$ for linear, isotropic material space, $\mathbf{H}$ and $\mathbf{B}$ are related according to $\mathbf{B} = \mu\mathbf{H}$. Fig 1.1 further shows the analogy between electric and magnetic field quantities. Some of the magnetic field quantities will be introduced later in this chapter, and others will be presented in the next. The analogy is presented here to show that most of the equations we have derived for the electric fields may be readily used to obtain corresponding equations for magnetic fields if the equivalent analogous quantities are substituted. This way it does not appear as if we are learning new concepts. A definite link between electric and magnetic fields was established by Oersted in 1820. As we have noticed, an electrostatic field is produced by static or stationary charges. If the charges are moving with constant velocity, a static magnetic (or magnetostatic) field is produced. A magnetostatic field is produced by a constant current flow (or direct current). This current flow may be due to magnetization currents as in permanent magnets, electron-beam currents in vacuum tubes, or conduction currents as in current-carrying wires. In this chapter, we consider magnetic fields in free space due to direct current. Our study of magnetostatics is not a dispensable luxury but an indispensable necessity. Motors, transformers, microphones, compasses, telephone bell ringers, television focusing controls, advertising displays, magnetically levitated high-speed vehicles, memory stores, magnetic separators, and so on, which play an important role in our everyday life, could not have been developed without an understanding of magnetic phenomena. There are two major laws governing magnetostatic fields: (1) Biot–Savart’s law, and (2) Ampère’s circuit law. Like Coulomb’s law, Biot–Savart’s law is the general law of magnetostatics. Just as Gauss’s law is a special case of Coulomb’s law, Ampère’s law is a special

case of Biot–Savart’s law and is easily applied in problems involving symmetrical current distribution. The two laws of magnetostatics are stated and applied first, with their derivations provided later in the chapter.

Term	Electric	Magnetic
Basic laws	$\mathbf{F} = \frac{Q_1 Q_2}{4\pi\epsilon R^2} \mathbf{a}_R$ $\oint \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}}$	$d\mathbf{B} = \frac{\mu_0 I d\mathbf{l} \times \mathbf{a}_R}{4\pi R^2}$ $\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$
Force law	$\mathbf{F} = Q\mathbf{E}$	$\mathbf{F} = Q\mathbf{u} \times \mathbf{B}$
Source element	dQ	$dQ\mathbf{u} = Id\mathbf{l}$
Field intensity	$E = \frac{V}{\ell} \text{ (V/m)}$	$H = \frac{I}{\ell} \text{ (A/m)}$
Flux density	$D = \frac{\psi}{S} \text{ (C/m}^2\text{)}$	$B = \frac{\psi}{S} \text{ (Wb/m}^2\text{)}$
Relationship between fields	$\mathbf{D} = \epsilon\mathbf{E}$	$\mathbf{B} = \mu\mathbf{H}$
Potentials	$E = -\nabla V$ $V = \int_L \frac{\rho_L d\ell}{4\pi\epsilon R}$	$\mathbf{H} = -\nabla V_m \text{ (J = 0)}$ $\mathbf{A} = \int_L \frac{\mu I d\ell}{4\pi R}$
Flux	$\psi = \int \mathbf{D} \cdot d\mathbf{S}$ $\psi = Q = CV$ $I = C \frac{dV}{dt}$	$\psi = \int_s \mathbf{B} \cdot d\mathbf{S}$ $\psi = LI$ $V = L \frac{dI}{dt}$
Energy density	$w_E = \frac{1}{2} \mathbf{D} \cdot \mathbf{E}$	$w_m = \frac{1}{2} \mathbf{B} \cdot \mathbf{H}$
Poisson’s equation	$\nabla^2 V = -\frac{\rho_v}{\epsilon}$	$\nabla^2 \mathbf{A} = -\mu\mathbf{J}$

Figure 1.1: Analogy between Electric and Magnetic Fields

Biot–Savart’s Law

Biot–Savart’s law states that the differential magnetic field intensity $d\mathbf{H}$ produced at a point P , as shown in Figure 1.2, by the differential current element $I d\mathbf{l}$ is proportional to the product $I d\mathbf{l}$ and the sine of the angle α between the element and the line joining P to the element and is inversely proportional to the square of the

distance R between P and the element. That is,

$$dH \propto \frac{I dl \sin \alpha}{R^2} \quad (1.1)$$

or

$$dH = k \frac{I dl \sin \alpha}{R^2} \quad (1.2)$$

where k is the constant of proportionality. In SI units, $k = \frac{1}{4\pi}$, so eq. (1.2) becomes

$$dH = \frac{I dl \sin \alpha}{4\pi R^2} \quad (1.3)$$

From the definition of cross product, it is easy to notice that eq. (1.3) is better put in vector form as

$$d\mathbf{H} = \frac{I d\mathbf{l} \times \mathbf{a}_R}{4\pi R^2} = \frac{I d\mathbf{l} \times \mathbf{R}}{4\pi R^3} \quad (1.4)$$

where $R = |\mathbf{R}|$ and $\mathbf{a}_R = \mathbf{R}/R$; $\mathbf{R}$ and $d\mathbf{l}$ are illustrated in Figure 1.2. Thus the direction of $d\mathbf{H}$ can be determined by the right-hand rule with the right-hand thumb pointing in the direction of the current and the right-hand fingers encircling the wire in the direction of $d\mathbf{H}$ as shown in Figure 1.3(a). Alternatively, we can use the right-hand-screw rule to determine the direction of $d\mathbf{H}$: with the screw placed along the wire and pointed in the direction of current flow, the direction of rotation of the screw is the direction of $d\mathbf{H}$ as in Figure 1.3(b).

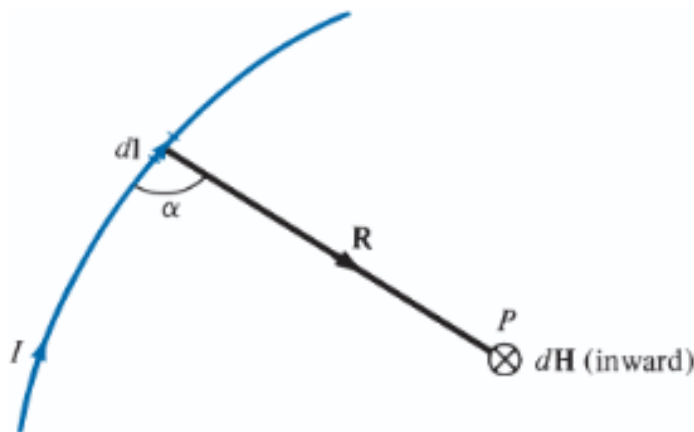


Figure 1.2: Magnetic field $d\mathbf{H}$ at P due to current element $I d\mathbf{l}$.

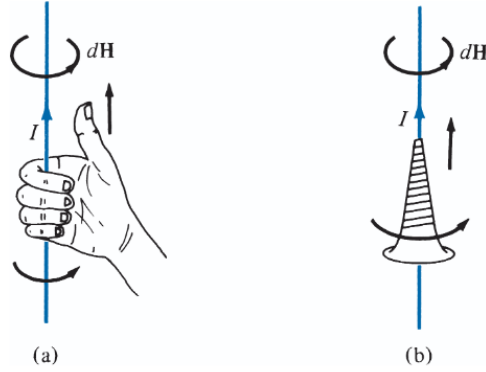


Figure 1.3: Determining the direction of $d\mathbf{H}$ using (a) the right-hand rule or (b) the right-handed-screw rule.

It is customary to represent the direction of the magnetic field intensity $\mathbf{H}$ (or current I) by a small circle with a dot or cross sign depending on whether $\mathbf{H}$ (or I) is out of the page, or into it respectively, as illustrated in Figure 1.4. Just as we can have different charge configurations, we can have different current distributions: line current, surface current, and volume current as shown in Figure 1.5. If we define $\mathbf{K}$ as the surface current density in amperes per meter and $\mathbf{J}$ as the volume current density in amperes per meter squared, the source elements are related as

$$I d\mathbf{l} = \mathbf{K} d\mathbf{S} = \mathbf{J} dv \quad (1.5)$$

Thus in terms of the distributed current sources, the Biot–Savart’s law as in eq. (1.4) becomes

$$\mathbf{H} = \int_L \frac{I d\mathbf{l} \times \mathbf{a}_R}{4\pi R^2} \quad (\text{line current}) \quad (1.6)$$

$$\mathbf{H} = \int_S \frac{\mathbf{K} d\mathbf{S} \times \mathbf{a}_R}{4\pi R^2} \quad (\text{surface current}) \quad (1.7)$$

$$\mathbf{H} = \int_V \frac{\mathbf{J} dv \times \mathbf{a}_R}{4\pi R^2} \quad (\text{volume current}) \quad (1.8)$$

where $\mathbf{a}_R$ is a unit vector pointing from the differential element of current to the point of interest. As an example, let us apply eq. (1.6) to determine the field due to a straight current-carrying filamentary conductor of finite length AB as in Figure 1.5. We assume that the conductor is along the z -axis with its upper and lower ends, respectively, subtending angles α_2 and α_1 at P , the point at which $\mathbf{H}$ is to be

determined. Note that current flows from point A , where $\alpha = \alpha_1$, to point B , where $\alpha = \alpha_2$. If we consider the contribution $d\mathbf{H}$ at P due to an element $d\mathbf{l}$ at $(0, 0, z)$,

$$d\mathbf{H} = \frac{I d\mathbf{l} \times \mathbf{R}}{4\pi R^3} \quad (1.9)$$

But $d\mathbf{l} = dz \mathbf{a}_z$ and $\mathbf{R} = \rho \mathbf{a}_\rho - z \mathbf{a}_z$, so

$$d\mathbf{l} \times \mathbf{R} = \rho dz \mathbf{a}_\phi \quad (1.10)$$

Hence,

$$\mathbf{H} = \int \frac{I \rho dz}{4\pi(\rho^2 + z^2)^{3/2}} \mathbf{a}_\phi \quad (1.11)$$

Letting $z = \rho \cot \alpha$, $dz = -\rho \csc^2 \alpha d\alpha$, $(\rho^2 + z^2)^{3/2} = \rho^3 \csc^3 \alpha$, and eq. (1.11) becomes

$$\mathbf{H} = -\frac{1}{4\pi} \int_{\alpha_1}^{\alpha_2} \frac{\rho^2 \csc^2 \alpha d\alpha}{\rho^3 \csc^3 \alpha} \mathbf{a}_\phi \quad (1.12)$$

$$= -\frac{I}{4\pi\rho} \mathbf{a}_\phi \int_{\alpha_1}^{\alpha_2} \sin \alpha d\alpha \quad (1.13)$$

or

$$\mathbf{H} = \frac{I}{4\pi\rho} (\cos \alpha_2 - \cos \alpha_1) \mathbf{a}_\phi \quad (1.14)$$

This expression is generally applicable for any straight filamentary conductor. For a semi-infinite conductor:

$$\mathbf{H} = \frac{I}{4\pi\rho} \mathbf{a}_\phi \quad (1.15)$$

For an infinite conductor:

$$\mathbf{H} = \frac{I}{2\pi\rho} \mathbf{a}_\phi \quad (1.16)$$

To find the unit vector $\mathbf{a}_\phi$ in eqs. (1.12)–(1.14), a simple approach is

$$\mathbf{a}_\phi = \mathbf{a}_\ell \times \mathbf{a}_\rho \quad (1.17)$$

where $\mathbf{a}_\ell$ is a unit vector along the line current and $\mathbf{a}_\rho$ is a unit vector along the perpendicular line from the current to the field point.

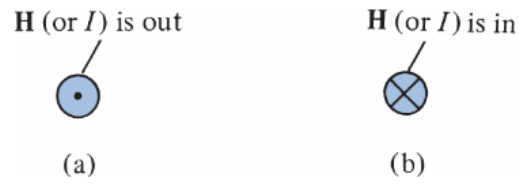


Figure 1.4: Conventional representation of $\mathbf{H}$ (or $\mathbf{I}$) (a) out of the page and (b) into the page

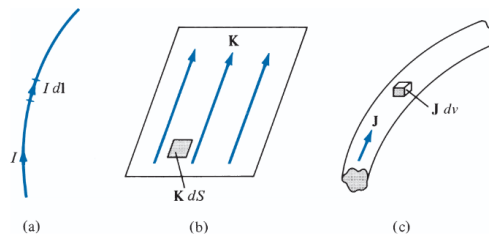


Figure 1.5: Current distributions: (a) line current, (b) surface current, (c) volume current

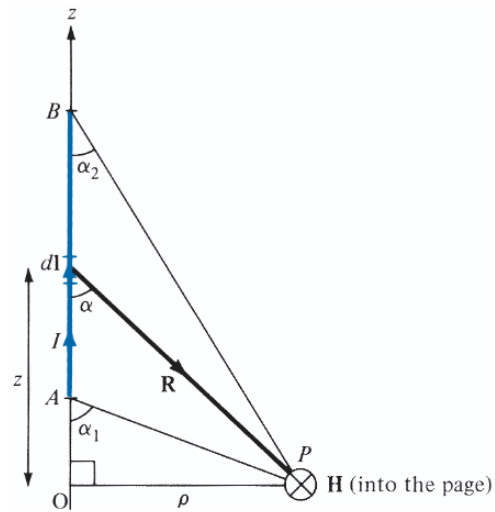


Figure 1.6: Field at point P due to a straight filamentary conductor.

Example 1-Circular Loop: A circular loop located on $x^2 + y^2 = 9$, $z = 0$ carries a direct current of 10 A along $\mathbf{a}_\phi$. Determine $\mathbf{H}$ at $(0, 0, 4)$ and $(0, 0, -4)$.

Solution

Consider the circular loop shown in Figure 1.7 (a). The magnetic field intensity $d\mathbf{H}$ at point $P(0, 0, h)$ contributed by current element $I d\mathbf{l}$ is given by Biot–Savart’s law:

$$d\mathbf{H} = \frac{I d\mathbf{l} \times \mathbf{R}}{4\pi R^3}$$

where

$$d\mathbf{l} = \rho d\phi \mathbf{a}_\phi, \quad \mathbf{R} = (0, 0, h) - (x, y, 0) = -\rho \mathbf{a}_\rho + h \mathbf{a}_z$$

and

$$d\mathbf{l} \times \mathbf{R} = \begin{vmatrix} \mathbf{a}_\rho & \mathbf{a}_\phi & \mathbf{a}_z \\ 0 & \rho d\phi & 0 \\ -\rho & 0 & h \end{vmatrix} = \rho h d\phi \mathbf{a}_\rho + \rho^2 d\phi \mathbf{a}_z$$

Hence,

$$\begin{aligned} d\mathbf{H} &= \frac{I}{4\pi(\rho^2 + h^2)^{3/2}} (\rho h d\phi \mathbf{a}_\rho + \rho^2 d\phi \mathbf{a}_z) \\ &= dH_\rho \mathbf{a}_\rho + dH_z \mathbf{a}_z \end{aligned}$$

By symmetry, the contributions along $\mathbf{a}_\rho$ add up to zero because the radial components produced by current element pairs 180° apart cancel. This may also be shown mathematically by writing $\mathbf{a}_\rho$ in rectangular coordinate system (i.e., $\mathbf{a}_\rho = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y$). Integrating $\cos \phi$ or $\sin \phi$ over $0 \leq \phi \leq 2\pi$ gives zero, thereby showing that $H_\rho = 0$. Thus

$$\mathbf{H} = \int dH_z \mathbf{a}_z = \int_0^{2\pi} \frac{I \rho^2 d\phi}{4\pi [\rho^2 + h^2]^{3/2}} \mathbf{a}_z = \frac{I \rho^2 2\pi}{4\pi [\rho^2 + h^2]^{3/2}} \mathbf{a}_z$$

or

$$\mathbf{H} = \frac{I \rho^2}{2 [\rho^2 + h^2]^{3/2}} \mathbf{a}_z$$

(a) Substituting $I = 10$ A, $\rho = 3$, $h = 4$ gives

$$\mathbf{H}(0, 0, 4) = \frac{10(3)^2}{2[9 + 16]^{3/2}} \mathbf{a}_z = 0.36 \mathbf{a}_z \text{ A/m}$$

(b) Notice from $d\mathbf{l} \times \mathbf{R}$ in the Biot–Savart law that if h is replaced by $-h$, the z -component of $d\mathbf{H}$ remains the same while the ρ -component still adds up to zero due to the axial symmetry of the loop. Hence

$$\mathbf{H}(0, 0, -4) = \mathbf{H}(0, 0, 4) = 0.36 \mathbf{a}_z \text{ A/m}$$

The flux lines due to the circular current loop are sketched in Figure 1.7 (b).

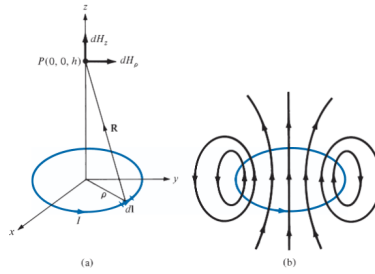


Figure 1.7: Fig 1.7: (a) circular current loop, (b) flux lines due to the current loop

Example 2: Solenoid

A solenoid of length ℓ and radius a consists of N turns of wire carrying current I . Show that at point P along its axis,

$$\mathbf{H} = \frac{nI}{2}(\cos \theta_2 - \cos \theta_1) \mathbf{a}_z$$

where $n = N/\ell$, θ_1 and θ_2 are the angles subtended at P by the end turns as illustrated in Figure 1.8. Also show that if $\ell \gg a$, at the center of the solenoid,

$$\mathbf{H} = nI \mathbf{a}_z$$

Solution: Consider the cross section of the solenoid as shown in Figure 1.8. Since the solenoid consists of circular loops, we apply the result of Example 1. The contribution to the magnetic field $\mathbf{H}$ at P by an element of the solenoid of length dz is

$$dH_z = \frac{I dl a^2}{2[a^2 + z^2]^{3/2}} = \frac{I a^2 n dz}{2[a^2 + z^2]^{3/2}}$$

where $dl = n dz = (N/\ell) dz$. From Figure 1.7,

$$\tan \theta = \frac{a}{z}; \quad \text{that is,}$$

$$dz = -a \csc^2 \theta d\theta = -\frac{[z^2 + a^2]^{3/2}}{a^2} \sin \theta d\theta$$

Hence,

$$dH_z = -\frac{nI}{2} \sin \theta d\theta$$

or

$$H_z = -\frac{nI}{2} \int_{\theta_1}^{\theta_2} \sin \theta d\theta$$

Thus

$$\mathbf{H} = \frac{nI}{2} (\cos \theta_2 - \cos \theta_1) \mathbf{a}_z$$

as required. Substituting $n = N/\ell$ gives

$$\mathbf{H} = \frac{NI}{2\ell} (\cos \theta_2 - \cos \theta_1) \mathbf{a}_z$$

At the center of the solenoid,

$$\cos \theta_2 = \frac{\ell/2}{\sqrt{a^2 + \ell^2/4}} = -\cos \theta_1$$

and

$$\mathbf{H} = \frac{In\ell}{2\sqrt{a^2 + \ell^2/4}} \mathbf{a}_z$$

If $\ell \gg a$ or $\theta_2 \approx 0^\circ$, $\theta_1 \approx 180^\circ$,

$$\mathbf{H} = nI \mathbf{a}_z = \frac{NI}{\ell} \mathbf{a}_z$$

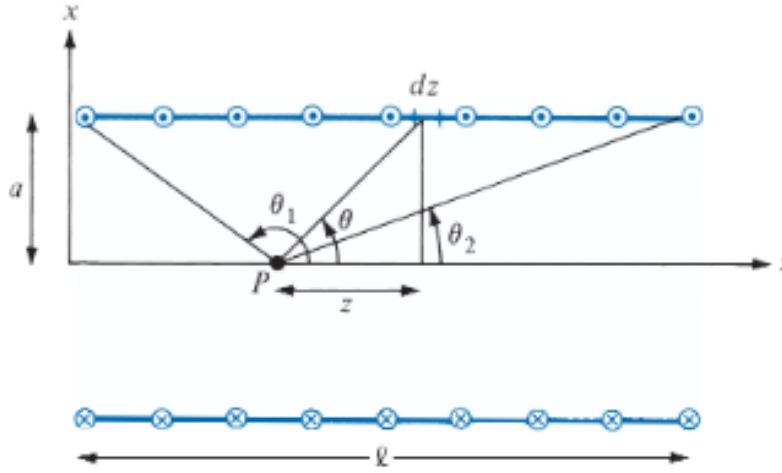


Figure 1.8: For Example 2; cross section of a solenoid

Ampère's Circuit Law—Maxwell's Equation

Ampère's circuit law states that the line integral of $\mathbf{H}$ around a closed path is the same as the net current I_{enc} enclosed by the path. In other words, the circulation of $\mathbf{H}$ equals I_{enc} ; that is,

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} \quad (1.18)$$

Ampère's law is similar to Gauss's law, since Ampère's law is easily applied to determine $\mathbf{H}$ when the current distribution is symmetrical. It should be noted that eq. (1.18) always holds regardless of whether the current distribution is symmetrical or not, but we can use the equation to determine $\mathbf{H}$ only when a symmetrical current distribution exists. Ampère's law is a special case of Biot–Savart's law; the former may be derived from the latter. By applying Stokes's theorem to the left-hand side of eq. (1.18), we obtain

$$I_{\text{enc}} = \oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} \quad (1.19)$$

But

$$I_{\text{enc}} = \int_S \mathbf{J} \cdot d\mathbf{S} \quad (1.20)$$

Comparing the surface integrals in eqs. (1.19) and (1.20) clearly reveals that

$$\nabla \times \mathbf{H} = \mathbf{J} \quad (1.21)$$

This is the third Maxwell's equation to be derived; it is essentially Ampère's law in differential (point) form, whereas eq. (1.18) is the integral form. From eq. (1.21), we should observe that $\nabla \times \mathbf{H} = \mathbf{J} \neq 0$; that is, a magnetostatic field is not conservative.

Applications of Ampère's Law

We now apply Ampère's circuit law to determine $\mathbf{H}$ for some symmetrical current distributions as we did for Gauss's law. We will consider an infinite line current, an infinite sheet of current, and an infinitely long coaxial transmission line. In each case, we apply

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}.$$

For symmetrical current distribution, $\mathbf{H}$ is either parallel or perpendicular to $d\mathbf{l}$. When $\mathbf{H}$ is parallel to $d\mathbf{l}$, $|\mathbf{H}| = \text{constant}$.

A. Infinite Line Current

Consider an infinitely long filamentary current I along the z -axis as in Figure 1.9. To determine $\mathbf{H}$ at an observation point P , we allow a closed path to pass through P . This path, on which Ampère's law is to be applied, is known as an Amperian path. We choose a concentric circle as the Amperian path in view of eq. (1.16), which shows that $\mathbf{H}$ is constant provided ρ is constant. Since this path encloses the whole current I , according to Ampère's law,

$$I = \oint_L H_\phi \mathbf{a}_\phi \cdot \rho d\phi \mathbf{a}_\phi = H_\phi \oint_L \rho d\phi = H_\phi 2\pi\rho \quad (1.22)$$

or

$$\mathbf{H} = \frac{I}{2\pi\rho} \mathbf{a}_\phi \quad (1.23)$$

as expected from eq. (1.16).

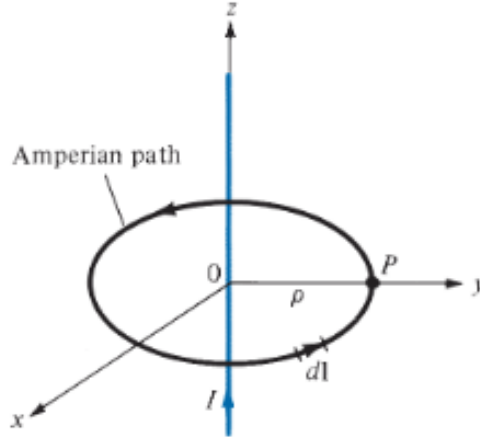


Figure 1.9: Ampère's law applied to an infinite filamentary line current.

B. Infinite Sheet of Current

Consider an infinite current sheet in the $z = 0$ plane. If the sheet has a uniform current density $\mathbf{K} = K_y \mathbf{a}_y$ A/m as shown in Figure 1.10, applying Ampère's law to the rectangular closed path 1-2-3-4-1 (Amperian path) gives

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} = K_y b \quad (1.24)$$

To evaluate the integral, we first need to have an idea of what $\mathbf{H}$ is like. To achieve this, we regard the infinite sheet as comprising filaments dH above and below the sheet due to a pair of filamentary currents can be found by using eqs. (1.16) and (1.17). As evident in Figure 1.10(b) The resultant dH has only an x -component. Also, $\mathbf{H}$ on one side of the sheet is the negative of that on the other side. Thus,

$$\mathbf{H} = \begin{cases} H_0 \mathbf{a}_x, & z > 0 \\ -H_0 \mathbf{a}_x, & z < 0 \end{cases} \quad (1.25)$$

where H_o is yet to be determined. Evaluating the line integral of $\mathbf{H}$ in eq. (1.25) along the closed path in Figure 1.10(a) gives

$$\begin{aligned}
\oint \mathbf{H} \cdot d\mathbf{l} &= \left(\int_1^2 + \int_2^3 + \int_3^4 + \int_4^1 \right) \mathbf{H} \cdot d\mathbf{l} \\
&= 0(-a) + (-H_o)(-b) + 0(a) + H_o(b) \\
&= 2H_o b
\end{aligned} \tag{1.26}$$

From eqs. (1.24) and (1.26), we obtain

$$H_o = \frac{1}{2} K_y.$$

Substituting H_o in eq. (1.25) gives

$$\mathbf{H} = \begin{cases} \frac{1}{2} K_y \mathbf{a}_x, & z > 0 \\ -\frac{1}{2} K_y \mathbf{a}_x, & z < 0 \end{cases} \tag{1.27}$$

In general, for an infinite sheet of current density $\mathbf{K}$ A/m,

$$\mathbf{H} = \frac{1}{2} \mathbf{K} \times \mathbf{a}_n \tag{1.28}$$

where $\mathbf{a}_n$ is a unit normal vector directed from the current sheet to the point of interest.

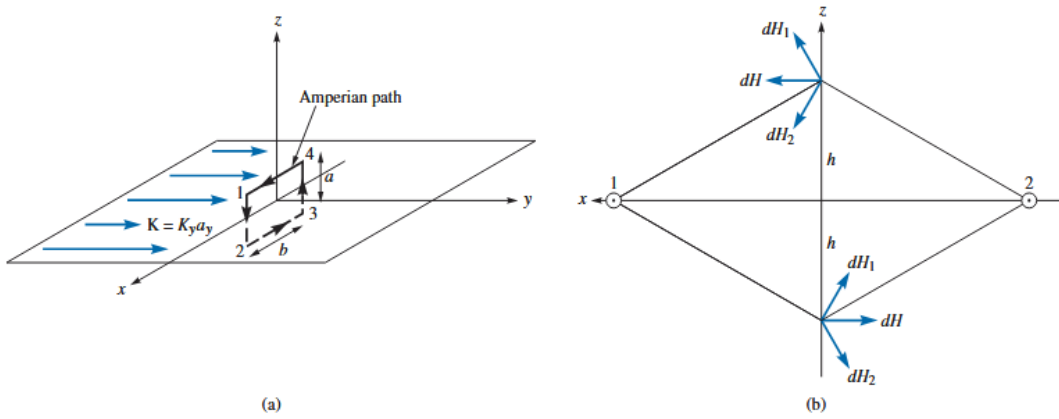


Figure 1.10: Application of Ampère's law to an infinite sheet: (a) closed path 1-2-3-4-1, (b) symmetrical pair of current filaments with current along y .

C. Infinitely Long Coaxial Transmission Line

Consider an infinitely long transmission line consisting of two concentric cylinders having their axes along the z -axis. The cross section of the line is shown in Figure 1.11, where the z -axis is out of the page. The inner conductor has radius a and carries current I , while the outer conductor has inner radius b and thickness t and carries return current $-I$. We want to determine $\mathbf{H}$ everywhere, assuming that current is uniformly distributed in both conductors. Since the current distribution is symmetrical, we apply Ampère's law along the Amperian path for each of the four possible regions: $0 \leq \rho \leq a$, $a \leq \rho \leq b$, $b \leq \rho \leq b + t$, and $\rho \geq b + t$. For region $0 \leq \rho \leq a$, we apply Ampère's law to path L_1 , giving

$$\oint_{L_1} \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} = \int_S \mathbf{J} \cdot d\mathbf{S} \quad (1.29)$$

Since the current is uniformly distributed over the cross section,

$$\mathbf{J} = \frac{I}{\pi a^2} \mathbf{a}_z, \quad d\mathbf{S} = \rho d\phi d\rho \mathbf{a}_z$$

$$I_{\text{enc}} = \int_S \mathbf{J} \cdot d\mathbf{S} = \frac{I}{\pi a^2} \int_0^{2\pi} \int_0^\rho \rho d\rho d\phi = \frac{I\rho^2}{a^2}$$

Hence eq. (1.29) becomes

$$H_\phi \oint_{L_1} dl = H_\phi 2\pi\rho = \frac{I\rho^2}{a^2}$$

or

$$H_\phi = \frac{I\rho}{2\pi a^2} \quad (1.30)$$

For region $a \leq \rho \leq b$, we use path L_2 as the Amperian path,

$$\oint_{L_2} \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} = I$$

$$H_\phi 2\pi\rho = I$$

or

$$H_\phi = \frac{I}{2\pi\rho} \quad (1.31)$$

For region $b \leq \rho \leq b + t$, we use path L_3 , getting

$$\oint_{L_3} \mathbf{H} \cdot d\mathbf{l} = H_\phi 2\pi\rho = I_{\text{enc}} \quad (1.32)$$

where

$$I_{\text{enc}} = I + \int_S \mathbf{J} \cdot d\mathbf{S}$$

and the current density in the outer conductor is

$$\mathbf{J} = -\frac{I}{\pi[(b+t)^2 - b^2]} \mathbf{a}_z$$

Thus

$$I_{\text{enc}} = I - \frac{I}{\pi[(b+t)^2 - b^2]} \int_0^{2\pi} \int_b^\rho \rho d\rho d\phi$$

$$I_{\text{enc}} = I \left[1 - \frac{\rho^2 - b^2}{t^2 + 2bt} \right]$$

Substituting into eq. (1.32),

$$H_\phi = \frac{I}{2\pi\rho} \left[1 - \frac{\rho^2 - b^2}{t^2 + 2bt} \right] \quad (1.33)$$

For region $\rho \geq b+t$, we use path L_4 , giving

$$\oint_{L_4} \mathbf{H} \cdot d\mathbf{l} = I - I = 0$$

or

$$H_\phi = 0 \quad (1.34)$$

Putting eqs. (1.31) to (1.34) together gives

$$\mathbf{H} = \begin{cases} \frac{I\rho}{2\pi a^2} \mathbf{a}_\phi, & 0 \leq \rho \leq a \\ \frac{I}{2\pi\rho} \mathbf{a}_\phi, & a \leq \rho \leq b \\ \frac{I}{2\pi\rho} \left[1 - \frac{\rho^2 - b^2}{t^2 + 2bt} \right] \mathbf{a}_\phi, & b \leq \rho \leq b+t \\ 0, & \rho \geq b+t \end{cases} \quad (1.35)$$

The magnitude of $\mathbf{H}$ is sketched in Figure 1.12. From these examples, it can be observed that the ability to take $\mathbf{H}$ from under the integral sign is the key to using Ampère's law to determine $\mathbf{H}$. In other words, Ampère's law can be used to find $\mathbf{H}$ only due to symmetric current distributions for which it is possible to find a closed path over which $\mathbf{H}$ is constant in magnitude.

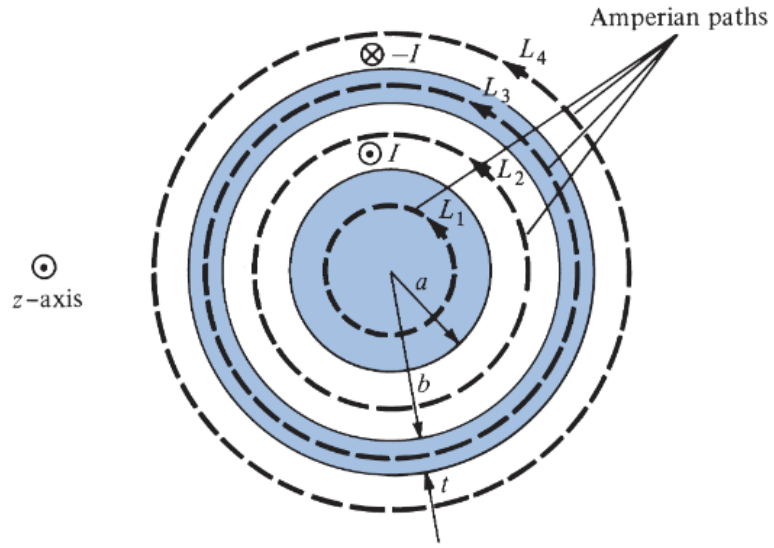


Figure 1.11: Cross section of the transmission line; the positive z -direction is out of the page

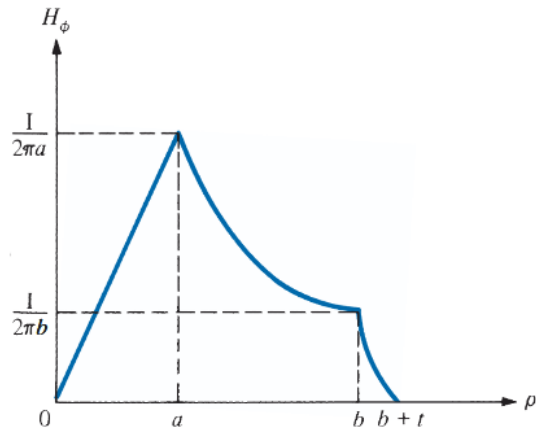


Figure 1.12: Plot of H_ϕ against ρ

7.7 Magnetic Scalar and Vector Potentials

We recall that some electrostatic field problems were simplified by relating the electric potential V to the electric field intensity $\mathbf{E}$ ($\mathbf{E} = -\nabla V$). Similarly, we can define a potential associated with magnetostatic field $\mathbf{B}$. In fact, the magnetic potential could be scalar V_m or vector $\mathbf{A}$. To define V_m and $\mathbf{A}$ involves recalling two important identities:

$$\nabla \times (\nabla V) = 0 \quad (1.36)$$

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0 \quad (1.37)$$

which must always hold for any scalar field V and vector field $\mathbf{A}$. Just as $\mathbf{E} = -\nabla V$, we define the magnetic scalar potential V_m (in amperes) as related to $\mathbf{H}$ according to

$$\mathbf{H} = -\nabla V_m \quad \text{if } \mathbf{J} = 0 \quad (1.38)$$

Combining eq. (1.38) and eq. (1.21) gives

$$\mathbf{J} = \nabla \times \mathbf{H} = \nabla \times (-\nabla V_m) = 0 \quad (1.39)$$

Thus the magnetic scalar potential V_m is only defined in a region where $\mathbf{J} = 0$. We also note that V_m satisfies Laplace's equation just as V does for electrostatic fields; hence,

$$\nabla^2 V_m = 0 \quad (\mathbf{J} = 0) \quad (1.40)$$

We know that for a magnetostatic field, $\nabla \cdot \mathbf{B} = 0$. To satisfy this simultaneously, we can define the magnetic vector potential $\mathbf{A}$ (in Wb/m) such that

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (1.41)$$

Just as we defined

$$V = \int \frac{dq}{4\pi\epsilon_0 R} \quad (1.42)$$

we can define

$$\mathbf{A} = \int_L \frac{\mu_0 I d\mathbf{l}}{4\pi R} \quad (\text{for line current}) \quad (1.43)$$

$$\mathbf{A} = \int_S \frac{\mu_0 \mathbf{K} dS}{4\pi R} \quad (\text{for surface current}) \quad (1.44)$$

$$\mathbf{A} = \int_V \frac{\mu_0 \mathbf{J} dv}{4\pi R} \quad (\text{for volume current}) \quad (1.45)$$

Rather than obtaining eqs. (1.43) to (1.45) from eq. (1.42), an alternative approach would be to obtain eqs. (1.43) to (1.45) from eqs. (1.6) to (1.8). For example, we can derive eq. (1.43) from eq. (1.6) in conjunction with eq. (1.41). To do this, we write eq. (1.6) as

$$\mathbf{B} = \frac{\mu_0}{4\pi} \int_L \frac{I d\mathbf{l}' \times \mathbf{R}}{R^3} \quad (1.46)$$

where $\mathbf{R}$ is the distance vector from the line element $d\mathbf{l}'$ at the source point (x', y', z') to the field point (x, y, z) , as shown in Figure 1.13 and $R = |\mathbf{R}|$; that is,

$$\mathbf{R} = \mathbf{r} - \mathbf{r}' = [(x - x')^2 + (y - y')^2 + (z - z')^2]^{1/2} \quad (1.47)$$

Hence,

$$\nabla \left(\frac{1}{R} \right) = -\frac{\mathbf{R}}{R^3} \quad (1.48)$$

Substituting this into eq. (1.46), we obtain

$$\mathbf{B} = -\frac{\mu_0}{4\pi} \int_L I d\mathbf{l}' \times \nabla \left(\frac{1}{R} \right) \quad (1.49)$$

We apply the vector identity

$$\nabla \times (f\mathbf{F}) = f\nabla \times \mathbf{F} + (\nabla f) \times \mathbf{F} \quad (1.50)$$

Taking $f = 1/R$ and $\mathbf{F} = d\mathbf{l}'$, we have

$$d\mathbf{l}' \times \nabla \left(\frac{1}{R} \right) = -\nabla \times \left(\frac{d\mathbf{l}'}{R} \right)$$

With this equation, eq. (1.49) reduces to

$$\mathbf{B} = \nabla \times \int_L \frac{\mu_0 I d\mathbf{l}'}{4\pi R} \quad (1.51)$$

Comparing eq. (1.51) with eq. (1.41) shows that

$$\mathbf{A} = \int_L \frac{\mu_0 I d\mathbf{l}'}{4\pi R}$$

verifying eq. (1.43). By substituting eq. (1.41) into eq. Magnetic flux and applying Stokes's theorem, we obtain

$$\Psi = \int_S \mathbf{B} \cdot d\mathbf{S} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \oint_L \mathbf{A} \cdot d\mathbf{l} \quad (1.52)$$

or

$$\Psi = \oint_L \mathbf{A} \cdot d\mathbf{l} \quad (1.53)$$

Thus the magnetic flux through a given area can be found by using either magnetic flux or (1.53). Also, the magnetic field can be determined by using either $\mathbf{V}_m$ or $\mathbf{A}$; the choice is dictated by the nature of the given problem except that V_m can be used only in a source-free region. The use of the magnetic vector potential provides a powerful, elegant approach to solving $\mathbf{H}$ problems, particularly those relating to antennas.

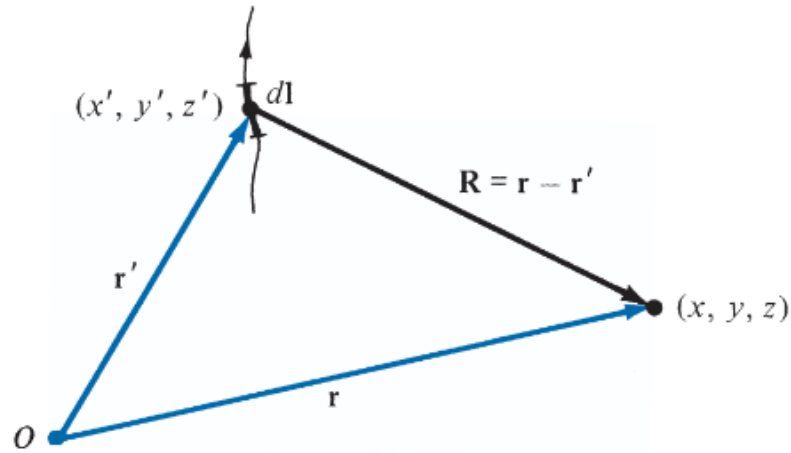


Figure 1.13: Illustration of the source point (x', y', z') and the field point (x, y, z) .

Derivation of Biot–Savart’s Law and Ampère’s Law

Both Biot–Savart’s law and Ampère’s law may be derived by using the concept of magnetic vector potential. The derivation will involve the use of the vector identities in eq. (1.50) and

$$\nabla \times \nabla \times \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \quad (1.54)$$

Since Biot–Savart’s law as given in eq. (1.4) is defined in terms of line current, we begin our derivation with eqs. (1.41) and (1.43); that is,

$$\mathbf{B} = \nabla \times \oint_L \frac{\mu_0 I d\mathbf{l}'}{4\pi R} = \frac{\mu_0 I}{4\pi} \oint_L \nabla \times \left(\frac{d\mathbf{l}'}{R} \right) \quad (1.55)$$

where R is as defined in eq. (1.47). If the vector identity in eq. (1.50) is applied by letting $\mathbf{F} = d\mathbf{l}'$ and $f = 1/R$, eq. (1.55) becomes

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \oint_L \left[\frac{1}{R} \nabla \times d\mathbf{l}' + \left(\nabla \frac{1}{R} \right) \times d\mathbf{l}' \right] \quad (1.56)$$

Since ∇ operates with respect to (x, y, z) and $d\mathbf{l}'$ is a function of (x', y', z') , $\nabla \times d\mathbf{l}' = 0$. Also

$$\frac{1}{R} = [(x - x')^2 + (y - y')^2 + (z - z')^2]^{-1/2} \quad (1.57)$$

$$\nabla \left(\frac{1}{R} \right) = - \frac{(x - x')\mathbf{a}_x + (y - y')\mathbf{a}_y + (z - z')\mathbf{a}_z}{[(x - x')^2 + (y - y')^2 + (z - z')^2]^{3/2}} = - \frac{\mathbf{a}_R}{R^2} \quad (1.58)$$

where $\mathbf{a}_R$ is a unit vector from the source point to the field point. Thus eq. (1.56) (upon dropping the prime in $d\mathbf{l}'$) becomes

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \oint_L \frac{d\mathbf{l} \times \mathbf{a}_R}{R^2} \quad (1.59)$$

which is Biot–Savart’s law. Using the identity in eq. (1.54) with eq. (1.41), we obtain

$$\nabla \times \mathbf{B} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \quad (1.60)$$

We choose

$$\nabla \cdot \mathbf{A} = 0 \quad (1.61)$$

which is called Coulomb’s gauge. Upon replacing $\mathbf{B}$ with $\mu_0 \mathbf{H}$ and using eq. (1.21), eq. (1.60) becomes

$$\nabla^2 \mathbf{A} = -\mu_0 \nabla \times \mathbf{H} \quad (1.62)$$

or

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J} \quad (1.63)$$

which is called the vector Poisson equation. It is similar to Poisson’s equation in electrostatics. In Cartesian coordinates, eq. (1.62) may be decomposed into three scalar equations:

$$\nabla^2 A_x = -\mu_0 J_x \quad (1.64)$$

$$\nabla^2 A_y = -\mu_0 J_y \quad (1.65)$$

$$\nabla^2 A_z = -\mu_0 J_z \quad (1.66)$$

which may be regarded as the scalar Poisson equations. It can also be shown that Ampère's circuit law is consistent with our definition of the magnetic vector potential. Using Stokes's theorem and eq. (1.41),

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} = \frac{1}{\mu_0} \int_S \nabla \times (\nabla \times \mathbf{A}) \cdot d\mathbf{S} \quad (1.67)$$

From eqs. (1.54), (1.61), and (1.62),

$$\nabla \times \nabla \times \mathbf{A} = -\nabla^2 \mathbf{A} = \mu_0 \mathbf{J}$$

Substituting this into eq. (1.67) yields

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{S} = I$$

which is Ampère's circuit law.

Magnetic Boundary Conditions

We define magnetic boundary conditions as the conditions that $\mathbf{H}$ (or $\mathbf{B}$) field must satisfy at the boundary between two different media. We make use of Gauss's law for magnetic fields

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0 \quad (1.68)$$

and Ampère's circuit law

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = I \quad (1.69)$$

Consider the boundary between two magnetic media 1 and 2, characterized, respectively, by μ_1 and μ_2 as in Figure 1.14. Applying eq. (1.68) to the pillbox (Gaussian surface) of Figure 1.14(a) and allowing $\Delta h \rightarrow 0$, we obtain

$$B_{1n} \Delta S - B_{2n} \Delta S = 0 \quad (1.70)$$

Thus

$$B_{1n} = B_{2n} \quad \text{or} \quad \mu_1 H_{1n} = \mu_2 H_{2n} \quad (1.71)$$

since $\mathbf{B} = \mu \mathbf{H}$. Equation (1.71) shows that the normal component of $\mathbf{B}$ is continuous at the boundary. It also shows that the normal component of $\mathbf{H}$ is discontinuous at the boundary; $\mathbf{H}$ undergoes some change at the interface. Similarly, we apply eq.

(1.69) to the closed path $abca$ of Figure 1.14(b), where surface current $\mathbf{K}$ on the boundary is assumed normal to the path. We obtain

$$K\Delta w = H_{1t}\Delta w + H_{1n}\frac{\Delta h}{2} + H_{2n}\frac{\Delta h}{2} - H_{2t}\Delta w - H_{2n}\frac{\Delta h}{2} - H_{1n}\frac{\Delta h}{2} \quad (1.72)$$

As $\Delta h \rightarrow 0$, eq. (1.72) leads to

$$H_{1t} - H_{2t} = K \quad (1.73)$$

This shows that the tangential component of $\mathbf{H}$ is also discontinuous. Equation (1.73) may be written in terms of $\mathbf{B}$ as

$$\frac{B_{1t}}{\mu_1} - \frac{B_{2t}}{\mu_2} = K \quad (1.74)$$

In the general case, eq. (1.73) becomes

$$(\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_{n12} = \mathbf{K} \quad (1.75)$$

where $\mathbf{a}_{n12}$ is a unit vector normal to the interface and is directed from medium 1 to medium 2. If the boundary is free of current or the media are not conductors (for $\mathbf{K}$ is free current density), $\mathbf{K} = 0$ and eq. (1.73) becomes

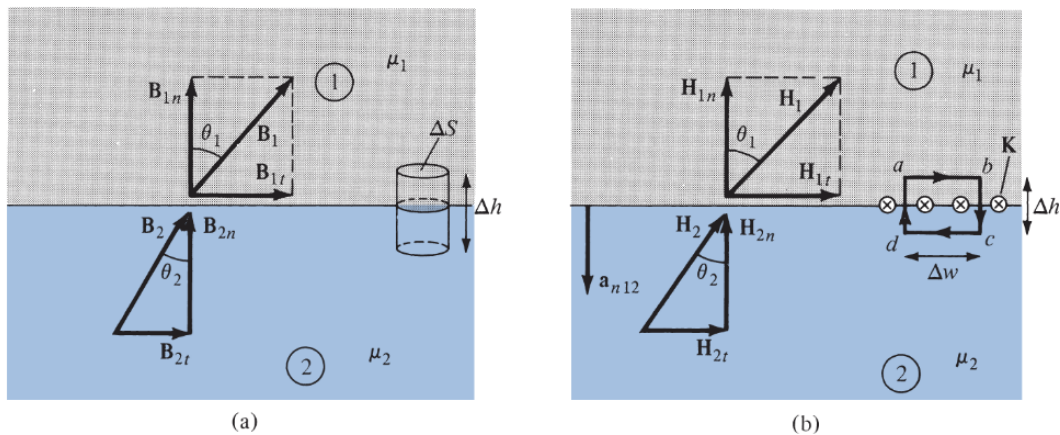


Figure 1.14: Boundary conditions between two magnetic media: (a) for $\mathbf{B}$, (b) for $\mathbf{H}$.

$$H_{1t} = H_{2t} \quad \text{or} \quad \frac{B_{1t}}{\mu_1} = \frac{B_{2t}}{\mu_2} \quad (1.76)$$

Thus the tangential component of $\mathbf{H}$ is continuous while that of $\mathbf{B}$ is discontinuous at the boundary. If the fields make an angle θ with the normal to the interface, eq. (1.71) results in

$$B_1 \cos \theta_1 = B_{1n} = B_{2n} = B_2 \cos \theta_2 \quad (1.77)$$

while eq. (1.76) produces

$$\frac{B_1}{\mu_1} \sin \theta_1 = H_{1t} = H_{2t} = \frac{B_2}{\mu_2} \sin \theta_2 \quad (1.78)$$

Dividing eq. (1.78) by eq. (1.77) gives

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2} \quad (1.79)$$

which is similar to the law of refraction for magnetic flux lines at a boundary with no surface current.